PAPER ID-310597

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2025-26
POWER SYSTEM PROTECTION

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	What are the types of protective relays?	1	K2
b.	What is protection zone?	1	K2
c.	What are the advantages of differential relay?	2	K2
d.	Why Buchholz relay is called gas actuated relay?	2	K2
e.	What do you understand by the term "Current Chopping"?	3	K2
f.	Give the classification of circuit breakers based on medium used for arc quenching.	4	K4
g.	Write the advantages of computer based protection system.	5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain need of power system protection. What are the different attributes of protection system? Explain in brief	1	K2
b.	Describe the operating principle, constructional features and area of applications distance relays or directional relay	2	K2
c.	What is carrier current protection? For what voltage range is it used for the protection of transmission lines?	3	K2
d.	Draw & describe the construction, working principle of vacuum circuit breaker.	4	K4
e.	What is level detector? Explain PNP and NPN transistor as level detector.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	What are basic requirements of protective relaying schemes?	1	K2
b.	What is a zone of protection? Discuss various zones of protection of a power system with the help of line diagram.	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Give a detailed comparison between static and electromagnetic relay.	2	K2
b.	Discuss the coincidence principle used in phase comparators.	2	K2

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	What do you understand by Carrier Current Protection scheme? Explain Phase Comparison Carrier Current Protection in detail.	3	K2
b.	Explain in detail about the operating principle of reverse power protection of alternator.	3	K2

6. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Discussion in detail the testing of the circuit breaker.	4	K4
b.	Discuss the properties of SF6 which makes it most suitable for circuit breakers.	4	K4

7. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the working principle of electronic relay and its advantages.	5	K2
b.	With the help of neat & clean diagram explain the microprocessor based digital static relay protection system.	5	K2

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